Dictionaries

→ Unordered → key-value pairs
→ keys should be ~~mut~~ immutable

(*) using the same key again overwrites the previous key's value.

(*) → get() method.

eg → a = {'student': 'Krish', 'grade': 'B'}

a.get('student') → returns 'Krish'

→ we can pass a default value in get(), if passed key is not found.

(*) eg. a.get('age', 'Unavailable')

→ ~~prints~~ returns unavailable.

(*) del a['grade']

deletes the key value pair of 'grade'.

(*) dict.keys() → returns all keys in the dict

(*) dict.values() → returns all the values in the dict.

(*) dict.items() → returns all the key-value pairs in the form of tuples.

Shallow copy (~~copy~~ dict.copy())

when we make a copy, ..
e.g. ~~a=~~ b = a.
and update a['name'] = 'Max'

then b['name'] also = 'Max'.

Note: mutable objects variables when copied * reference to the same object in memory.

Hence to make a copy,

```
b = a.copy()        # shallow copy
```

Dictionary comprehension

```
squares = {x : x**2 for x in range(5)}
```

=> {0:0, 1:1, 2:4, 3:9, 4:16}

We can use conditions,..

```
even_sq = {x : x**2 for x in range(4)
           if x%2==0}
```

=> {0:0, 2:4, 4:16}

Merge dictionaries

a = {'a':1} ; b = {'b':2}

```
m = {**a, **b}
```

=> m = {'a':1, 'b':2}

Note: **x
key value argument
(for any value present in form of key value pairs)